Mathematical Analysis. — On the Iteration of Rational Substitutions and Poincaré Functions. Note by M. S. Lattés.

Let $z_1 = R(z)$ be a rational map with distinct fixed points; let α be a fixed point with multiplier S, where $|S| > 1$. There is always at least one such point[1]. There then exists a function $\theta(u)$ which is meromorphic [or entire if $R(z)$ is a polynomial] satisfying, for all u, the functional equation:

$$\theta(Su) = R[\theta(u)] \tag{1}$$

and taking the value α at $u = 0$. The existence of $\theta(u)$, which we shall call the *Poincaré function* relative to the invariant point α, was established by Poincaré[2].

Poincaré, however, imposes on the desired function $\theta(u)$ the condition $\theta'(0) \neq 0$. If $\theta'(0)$ is zero and the first non-zero derivative for $u = 0$ is of order p, one still finds a meromorphic function $\theta(u)$ satisfying equation (1), but the number S appearing in this equation—which can still be called the multiplier—is related to $R'(\alpha)$ by the relation $S^p = R'(\alpha)$; the function $\theta(u)$ is then a meromorphic or entire function of u^p. For example, if the given substitution is $z_1 = 2z^2 - 1$, we can set $z = \cos u$, yielding $z_1 = \cos 2u$; here $\theta(u)$ is an entire function of u^2 [since $\cos u = 1 - \frac{u^2}{2} + \dots$] and we have $R'(\alpha) = 4 = S^2$ [where $\alpha = 1$, $R'(1) = 4$, and the multiplier is $S = 2$].

The Poincaré function $\theta(u)$ is nothing other than the inverse function of the Schröder function $\varphi(z)$, whose existence in the neighborhood of the point α was established by Mr. Kœ nigs and which satisfies the equation:

$$\varphi[R(z)] = S\varphi(z).$$

But, while the Poincaré function, when analytically continued from $u = 0$, gives rise to a single-valued (uniform) meromorphic or entire transcendental function, its inverse, the Schröder function, continued from $z = \alpha$, is by that very fact a multivalued (multiform) analytic function. Hence, there is an advantage to substituting the Poincaré function for the Schröder function if one wishes to study iteration starting from an arbitrary initial point z in the plane.

[1] Cf. Fatou, *Sur les substitutions rationnelles* (Comptes rendus, t. 165, 1917, p. 993).

[2] Poincaré, *Sur une classe nouvelle de transcendantes uniformes* (Journal de Mathématiques pures et appliquées, 1890).

For each value of z, the relation $z = \theta(u)$ yields at least one value of u, except for at most two exceptional values of z, according to Picard's Theorem. We then obtain, in parametric form, the values for the point z and its successive iterates [consequents] z_n:

$$z = \theta(u), \quad z_1 = \theta(Su), \quad z_2 = \theta(S^2 u), \quad \ldots, \quad z_n = \theta(S^n u).$$

These same formulas, for negative integer n, provide a sequence of successive antecedents [preimages] z_{-n} of the point z which, as $n \to \infty$, have a limit equal to $\theta(0)$, which is α. Thus, *one can choose, from among the possible successive preimages of any given point, a sequence of successive preimages z_{-n} tending, as $n \to \infty$, toward any of the invariant points α whose multiplier has a modulus greater than* 1.

The possible exceptional points α, β of the function $\theta(u)$ are either invariant points of the substitution[3] or two points forming a periodic cycle of order 2.

The problems relating to the iteration of a given substitution are thus transformed into problems concerning the growth of the function $\theta(u)$. To obtain categories of rational substitutions whose iteration is easy to study, it is convenient to choose substitutions that admit a Poincaré function belonging to the most common meromorphic (or entire) functions. Thus, if we assume S to be an integer, and $\theta(u)$ to be one of the functions $\tan u$, $\cos u$, $\wp(u)$ [the Weierstrass elliptic function], or a homographic transformation [Möbius transformation] of these functions, we obtain rational substitutions for which the method of parametric iteration allows us to completely resolve the fundamental problem of iteration. This problem, in its general form, can be stated as follows:

Determine the limit set E' (derived set) of the set E of iterates z_n of an arbitrarily given point z.

The set E' contains the iterates of its various points; on the other hand, we know that E', being a closed set, is the sum of a perfect set E'_1 and a countable set E'_2. The set E'_1 also contains the iterates of its various points, while for points in E'_2, their iterates may belong to E'_1. In the various examples listed above, we are able to completely determine E'_1 and E'_2, and determine what conditions z must satisfy for either of the sets E'_1 or E'_2 to be empty, or for a given point in the plane to belong to E'. These various conditions depend on the arithmetic properties of the number z.

[3]For example, for the substitution $z_1 = \frac{2z}{1-z^2}$, the Poincaré function is $z = \tan u$, with multiplier 2; this function admits the exceptional values i and $-i$, which are invariant points of the substitution.

Consider, for example, the substitution:

$$z_1 = \frac{(z^2+1)^2}{4z(z^2-1)} \tag{2}$$

which is obtained by setting:

$$z = \wp(u), \quad z_1 = \wp(2u) \quad \text{with} \quad g_2 = 4,\ g_3 = 0.$$

If 2ω denotes the real period, the other period is $2i\omega$. If we set:

$$z = \wp(u) = \wp(2\omega v + 2i\omega w),$$

where v and w are real, then to determine E', we must know the representations of v and w in the binary numeral system (base 2). The set E'_1 is formed by drawing the orthogonal grid of curves v = constant and w = constant (Cartesian ovals) and excluding from the plane the interior points of a countable infinity of quadrilaterals—contiguous or not—bounded by the curves of this grid. The common boundary between two excluded contiguous quadrilaterals must generally also be excluded. It is for the determination of the arcs of the ovals that bound the excluded domains that we must know the binary representations of v and w. We can thus completely solve the various iteration problems relating to (2).

The method of parametric iteration, using generalized Poincaré functions, can be applied in very broad cases to rational substitutions of two variables.